1 Image quality report

This document shows results for a total of 11461 image instances for the animal identification project pipeline.

1.1 Corrupt images

A total of 0 images were found to be corrupted and so unreadable. These images are not moved to the trusted zone.

1.2 Duplicate images

Duplicates are detected by computing an MD5 hash of each file's raw bytes: two images with the same hash are byte-for-byte identical. The first image seen is kept and later copies are flagged. A total of 5731 images were found to be duplicates of other images. Duplicates are not moved to the trusted zone.

Image
huggingface$animal-dataset$25-10-2026$train_None.jpg
huggingface$animal-dataset$27-10-2026$train_000000233339.jpg
huggingface$animal-dataset$27-10-2026$train_000000233379.jpg
huggingface$animal-dataset$27-10-2026$train_000000233383.jpg
huggingface$animal-dataset$27-10-2026$train_000000233435.jpg
huggingface$animal-dataset$27-10-2026$train_000000233439.jpg
huggingface$animal-dataset$27-10-2026$train_000000233446.jpg
huggingface$animal-dataset$27-10-2026$train_000000233500.jpg
huggingface$animal-dataset$27-10-2026$train_000000233520.jpg
huggingface$animal-dataset$27-10-2026$train_000000233538.jpg
huggingface$animal-dataset$27-10-2026$train_000000233553.jpg
huggingface$animal-dataset$27-10-2026$train_000000233560.jpg
huggingface$animal-dataset$27-10-2026$train_000000233566.jpg
huggingface$animal-dataset$27-10-2026$train_000000233578.jpg
huggingface$animal-dataset$27-10-2026$train_000000233613.jpg
huggingface$animal-dataset$27-10-2026$train_000000233637.jpg
huggingface$animal-dataset$27-10-2026$train_000000233647.jpg
huggingface$animal-dataset$27-10-2026$train_000000233660.jpg
huggingface$animal-dataset$27-10-2026$train_000000233786.jpg
huggingface$animal-dataset$27-10-2026$train_000000233815.jpg

Only the first 20 of 5731 images are listed.

1.3 Size outliers

To find outliers size is measured as the area (width x height) of each unique image. Using the interquartile range (IQR = Q3 - Q1), an image is an outlier when its area is below Q1 - 1.5 IQR (-923,088 px) or above Q3 + 1.5 IQR (1,672,880 px). A total of 1282 images were found to be size outliers.

Image	Width	Height	Area (px)
huggingface$animalimages$25-10-2026$train_cow_36.jpg	4458	3343	14,903,094
huggingface$animalimages$27-10-2026$train_cow_36.jpg	4458	3343	14,903,094
huggingface$animalimages$25-10-2026$train_pigeon_33.jpg	4669	3113	14,534,597
huggingface$animalimages$27-10-2026$train_pigeon_33.jpg	4669	3113	14,534,597
huggingface$animalimages$25-10-2026$train_hamster_38.jpg	4364	3273	14,283,372
huggingface$animalimages$27-10-2026$train_hamster_38.jpg	4364	3273	14,283,372
huggingface$animalimages$25-10-2026$train_hippopotamus_10.jpg	4340	3255	14,126,700
huggingface$animalimages$27-10-2026$train_hippopotamus_10.jpg	4340	3255	14,126,700

Image	Width	Height	Area (px)
huggingface$animalimages$25-10-2026$test_raccoon_37.jpg	3744	3744	14,017,536
huggingface$animalimages$27-10-2026$test_raccoon_37.jpg	3744	3744	14,017,536
huggingface$animalimages$25-10-2026$train_hyena_1.jpg	4320	3240	13,996,800
huggingface$animalimages$27-10-2026$train_hyena_1.jpg	4320	3240	13,996,800
huggingface$animalimages$25-10-2026$train_zebra_16.jpg	4410	3090	13,626,900
huggingface$animalimages$27-10-2026$train_zebra_16.jpg	4410	3090	13,626,900
huggingface$animalimages$25-10-2026$test_dragonfly_46.jpg	4896	2754	13,483,584
huggingface$animalimages$27-10-2026$test_dragonfly_46.jpg	4896	2754	13,483,584
huggingface$animalimages$25-10-2026$train_parrot_10.jpg	4224	3168	13,381,632
huggingface$animalimages$27-10-2026$train_parrot_10.jpg	4224	3168	13,381,632
huggingface$animalimages$25-10-2026$test_duck_7.jpg	4854	2742	13,309,668
huggingface$animalimages$27-10-2026$test_duck_7.jpg	4854	2742	13,309,668

Only the first 20 of 1282 images are listed.

1.4 Flat/blank images

To find flat images each image is converted to grayscale and the standard deviation of its pixel values is measured. A low standard deviation means the pixels barely change, so an image with a standard deviation below 5 (on a 0-255 scale) is considered flat: blank or a single solid colour. A total of 0 images were found to be flat.

1.5 Partly flat/blank images

Some images are only partly blank, e.g. a cropped or padded image with a solid border. To catch these, the grayscale image is split into four quadrants and the same standard deviation is measured in each one. An image that is not flat overall but has at least one quadrant below 5 is considered partly flat. This can also flag genuine photos with a uniform region such as a clear sky or a dark background, so the samples below should be checked. A total of 72 images were found to be partly flat.

Image	Lowest quadrant std. deviation
huggingface$animal-dataset$25-10-2026$train_000000246809.jpg	3.55
huggingface$animal-dataset$25-10-2026$train_000000250680.jpg	4.32
huggingface$animal-dataset$25-10-2026$train_000000255176.jpg	3.06
huggingface$animal-dataset$25-10-2026$train_000000256896.jpg	2.88
huggingface$animal-dataset$27-10-2026$train_000000246809.jpg	3.55
huggingface$animal-dataset$27-10-2026$train_000000250680.jpg	4.32
huggingface$animal-dataset$27-10-2026$train_000000255176.jpg	3.06
huggingface$animal-dataset$27-10-2026$train_000000256896.jpg	2.88
huggingface$animalimages$25-10-2026$test_butterfly_8.jpg	0.00
huggingface$animalimages$25-10-2026$test_crow_28.jpg	4.94
huggingface$animalimages$25-10-2026$test_hippopotamus_33.jpg	0.05
huggingface$animalimages$25-10-2026$test_rat_1.jpg	0.00
huggingface$animalimages$25-10-2026$test_sandpiper_49.jpg	2.93
huggingface$animalimages$25-10-2026$test_snake_58.jpg	4.87
huggingface$animalimages$25-10-2026$test_whale_2.jpg	3.61
huggingface$animalimages$25-10-2026$train_badger_59.jpg	0.00
huggingface$animalimages$25-10-2026$train_bat_21.jpg	0.51
huggingface$animalimages$25-10-2026$train_bat_26.jpg	4.31
huggingface$animalimages$25-10-2026$train_bat_37.jpg	0.94
huggingface$animalimages$25-10-2026$train_cockroach_43.jpg	3.78

Only the first 20 of 72 images are listed.

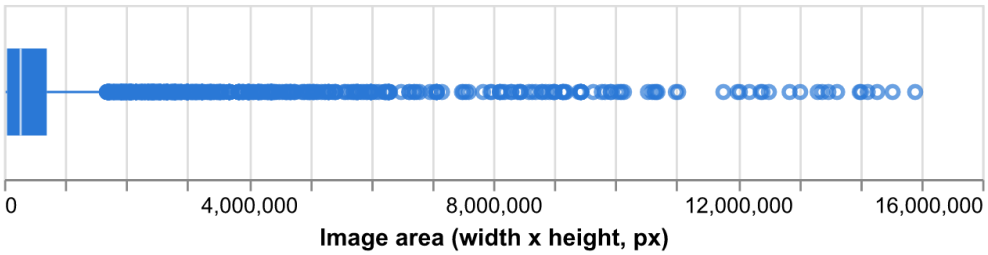


Figure 1: Distribution of image areas. Points beyond the whiskers are size outliers.



Figure 2: Sample partly flat images (flattest quadrant first).